AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 7 · WORKBOOK

Hypotheses and measurable outcomes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete hypothesis, outcome and revision record.

Keep direction, rubric, denominator, failure treatment, primary/secondary status and evidence limits together.

Keep one habit: Recall → W1–W3/W5 → independent primary metric → hypothesis project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

A literature comparison matrix

> **My remembered idea, example and correction**

Methods results and limitations

> **My remembered idea, example and correction**

A question that evidence can answer

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A hypothesis predicts a measurable comparison under stated conditions before inspecting results. Main primary prediction is B supported-answer rate>A on fixed paired cases, with support rubric and failure treatment declared. Supported completed answers/all planned cases is not the same as merely avoiding an unsupported claim: abstention and timeout do not count as supported.

  Preserve original plan when faster responses motivate a post-result speed claim; label exploratory revision and keep secondary outcomes separate. Added constructed six-case practice A/B both 4/6 supported; A has 1 unsupported/B has 1 abstention, both 1 timeout, mean final-outcome times 3/2 seconds. These are supplied records, not a live study.

  Separate picture ten-case design predicts fewer unsupported cases, defined by a factual claim absent from approved evidence. Friendly wording with a tied count does not meet that strict descriptive reduction. A constructed tie 2/2 is only practice, not actual reserved scoring. Equal/lower sample outcomes do not prove population equality; no power/significance result provided.

  Code order is not actual preregistration, and revealed practice is not fresh reserved evaluation.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Code order is not actual preregistration, and revealed practice is not fresh reserved evaluation.

- After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A hypothesis predicts a measurable comparison under stated conditions before inspecting results. Main primary prediction is B supported-answer rate>A on fixed paired cases, with support rubric and failure treatment declared. Supported completed answers/all planned cases is not the same as merely avoiding an unsupported claim: abstention and timeout do not count as supported.

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  Code order is not actual preregistration, and revealed practice is not fresh reserved evaluation.

- A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A hypothesis predicts a measurable comparison under stated conditions before inspecting results. Main primary prediction is B supported-answer rate>A on fixed paired cases, with support rubric and failure treatment declared. Supported completed answers/all planned cases is not the same as merely avoiding an unsupported claim: abstention and timeout do not count as supported.

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  Code order is not actual preregistration, and revealed practice is not fresh reserved evaluation.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled hypotheses and measurable outcomes **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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> | |
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Hypotheses and measurable outcomes. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Hypotheses and measurable outcomes. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

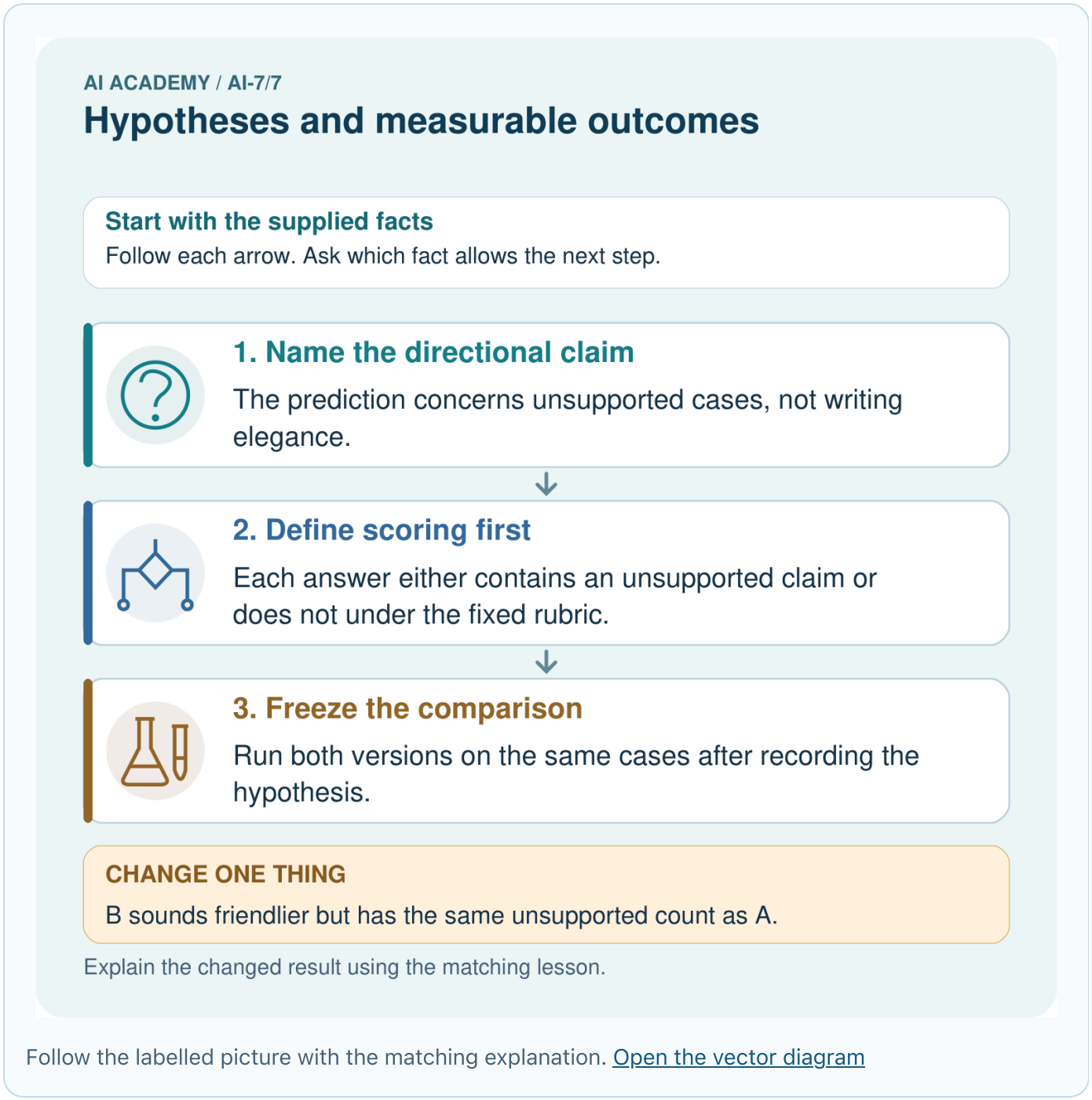
- Ten cases are reserved. Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

- Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.
- Which labelled step matches this detail? Run both versions on the same cases after recording the **hypothesis**.
- 1. Name the directional claim
- 2. Freeze the comparison

- 3. Define scoring first

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Freeze the comparison Run both versions on the same cases after recording the **hypothesis**. A **hypothesis** makes an expected pattern explicit before seeing outcomes.

The measured definition must be clear enough for another reviewer to apply.

Changed-case question and optional hint

- Changed-case practice: B sounds friendlier but has the same unsupported count as A. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. The stated improvement **hypothesis** is not supported on these cases.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The stated improvement **hypothesis** is not supported on these cases. Friendliness is a different outcome from the predefined unsupported-case count.

Repair and retain the record

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

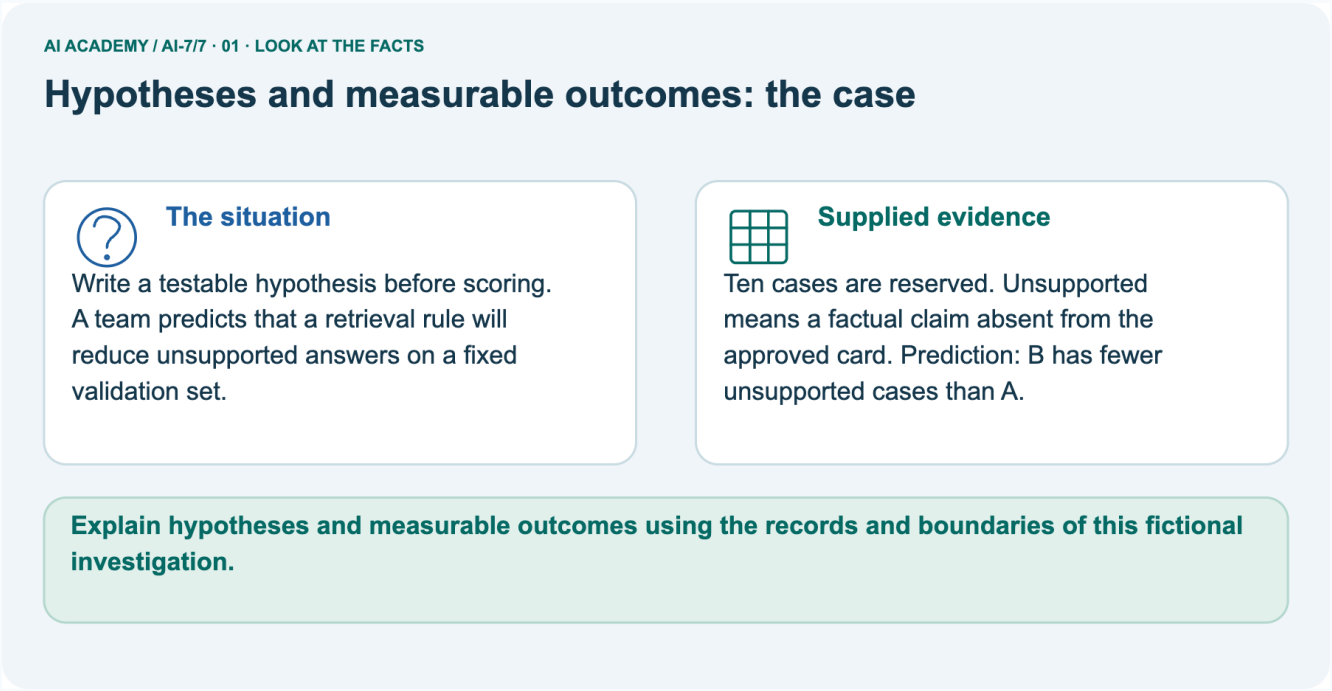
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Hypotheses and measurable outcomes: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

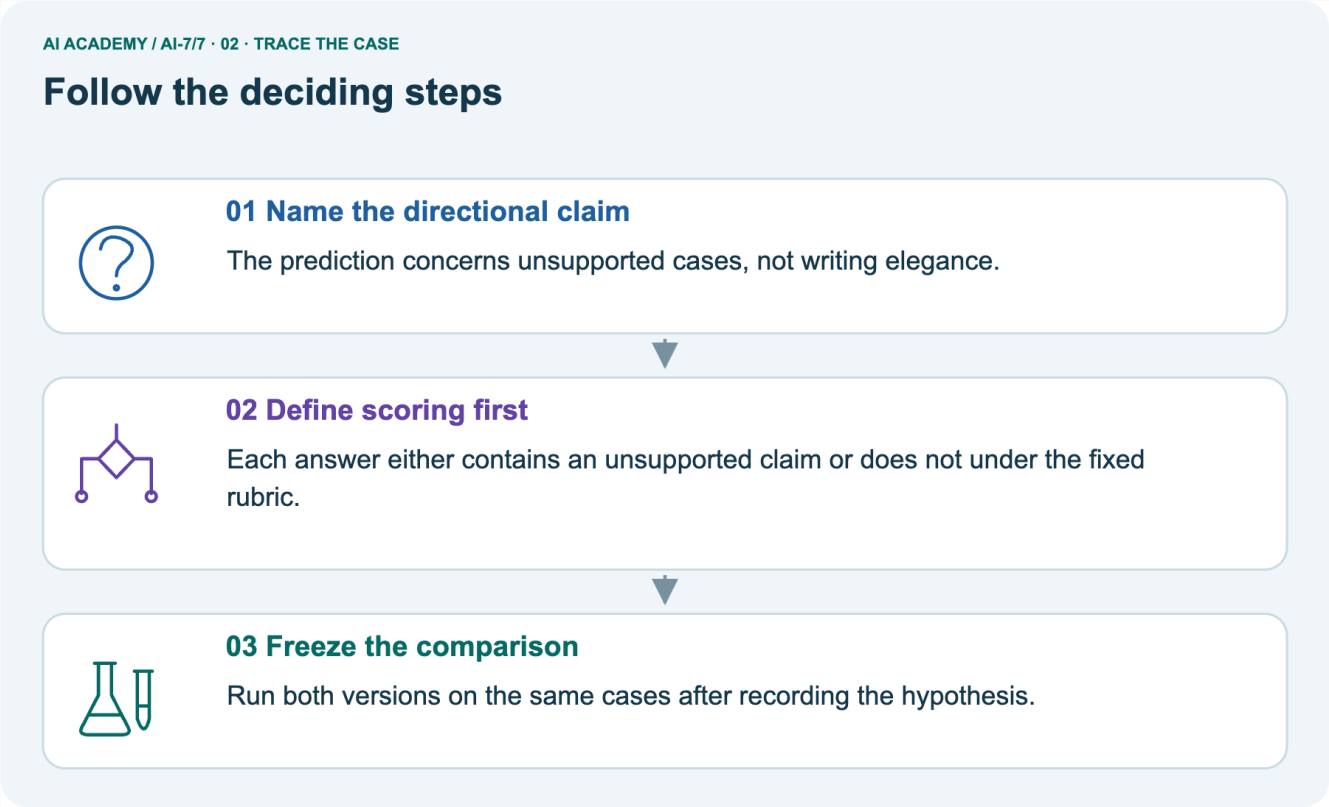
Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set. Supplied facts: Ten cases are reserved.

Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



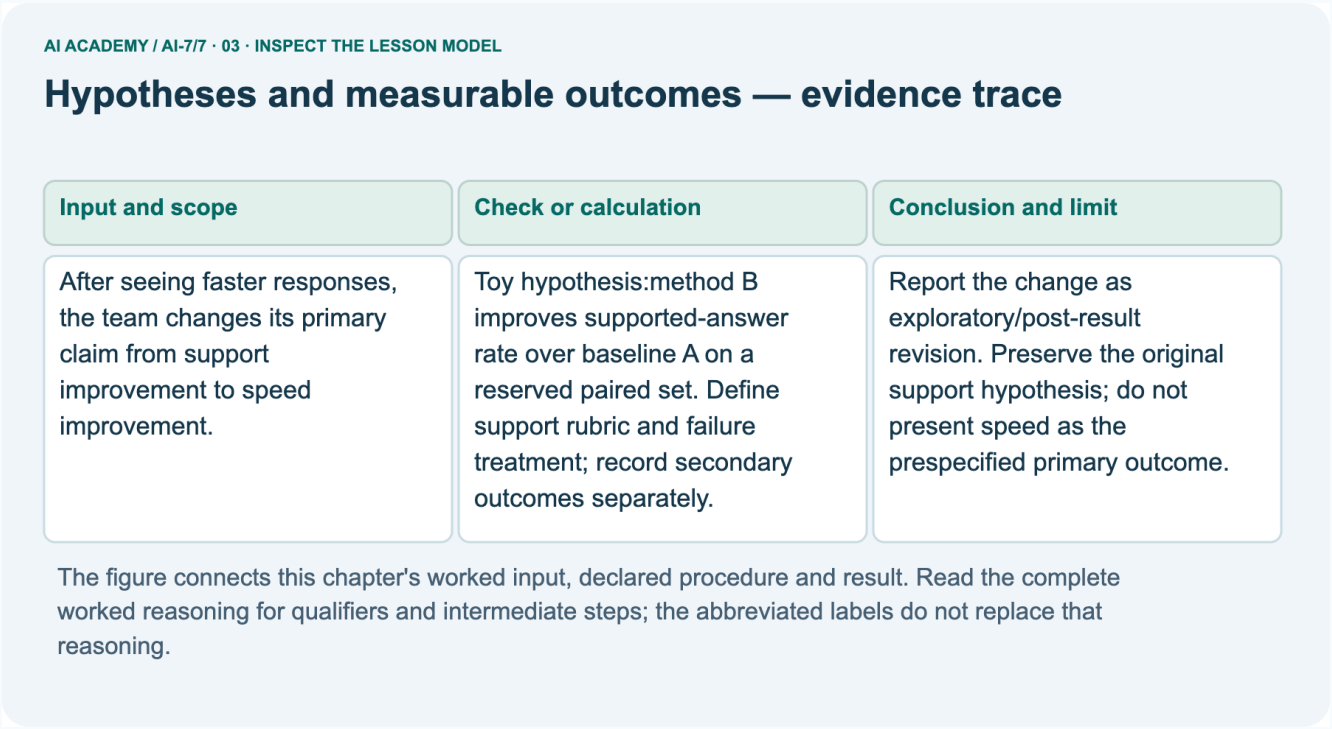
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the directional claim: The **prediction** concerns unsupported cases, not writing elegance. → Define scoring first: Each answer either contains an unsupported claim or does not under the fixed rubric. → Freeze the comparison: Run both versions on the same cases after recording the **hypothesis**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Hypotheses and measurable outcomes — evidence trace



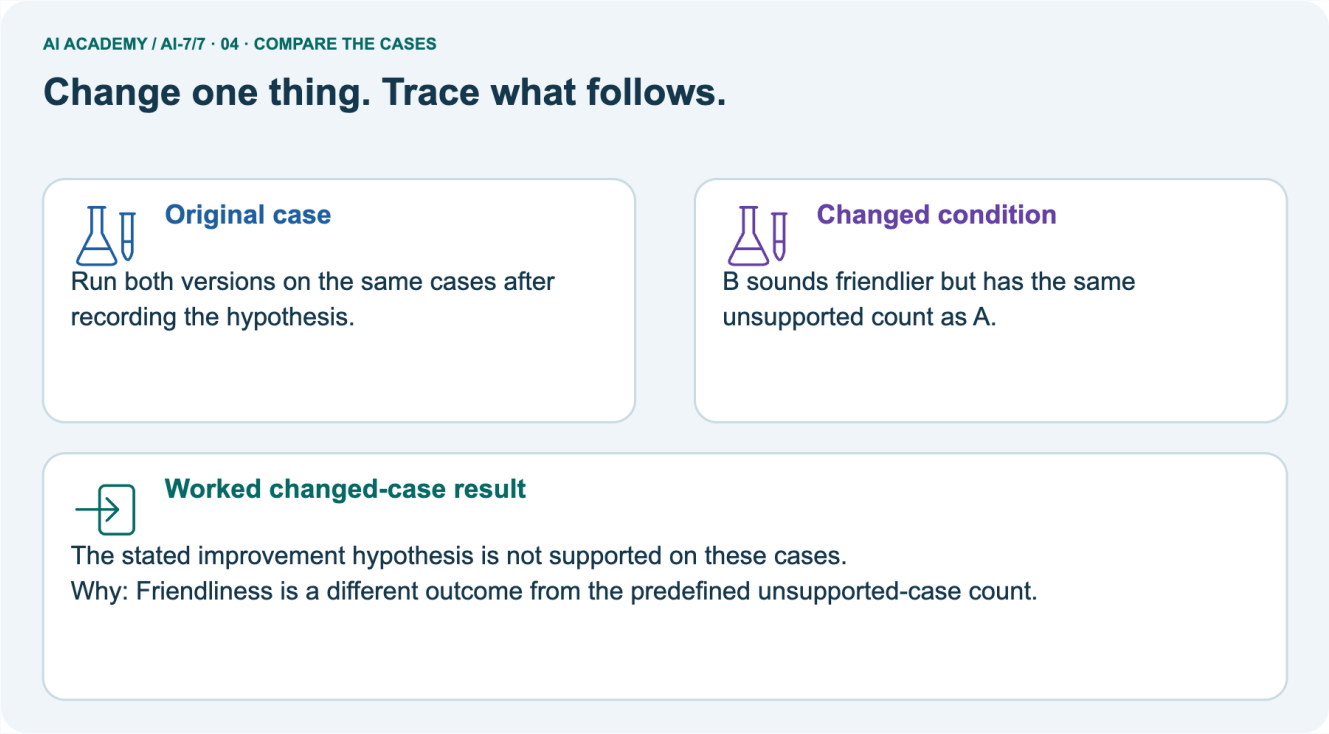
Follow the labelled picture with the matching explanation. Open the vector diagram

Hypotheses and measurable outcomes — **evidence** trace. After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. | Toy **hypothesis**: method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately. | Report the change as exploratory/post-result revision. Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

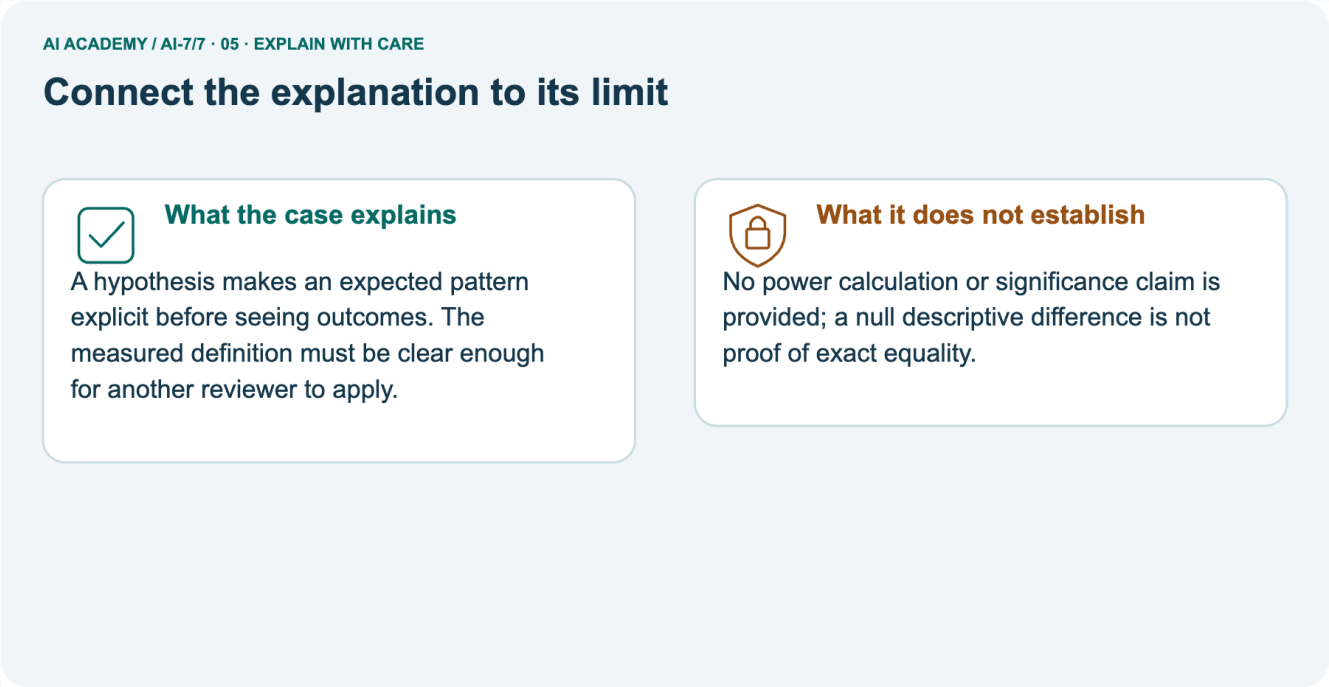
Original case: Run both versions on the same cases after recording the **hypothesis**. Changed condition: B sounds friendlier but has the same unsupported count as A.

Expected result: The stated improvement **hypothesis** is not supported on these cases.
Reason: Friendliness is a different outcome from the predefined unsupported-case count.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

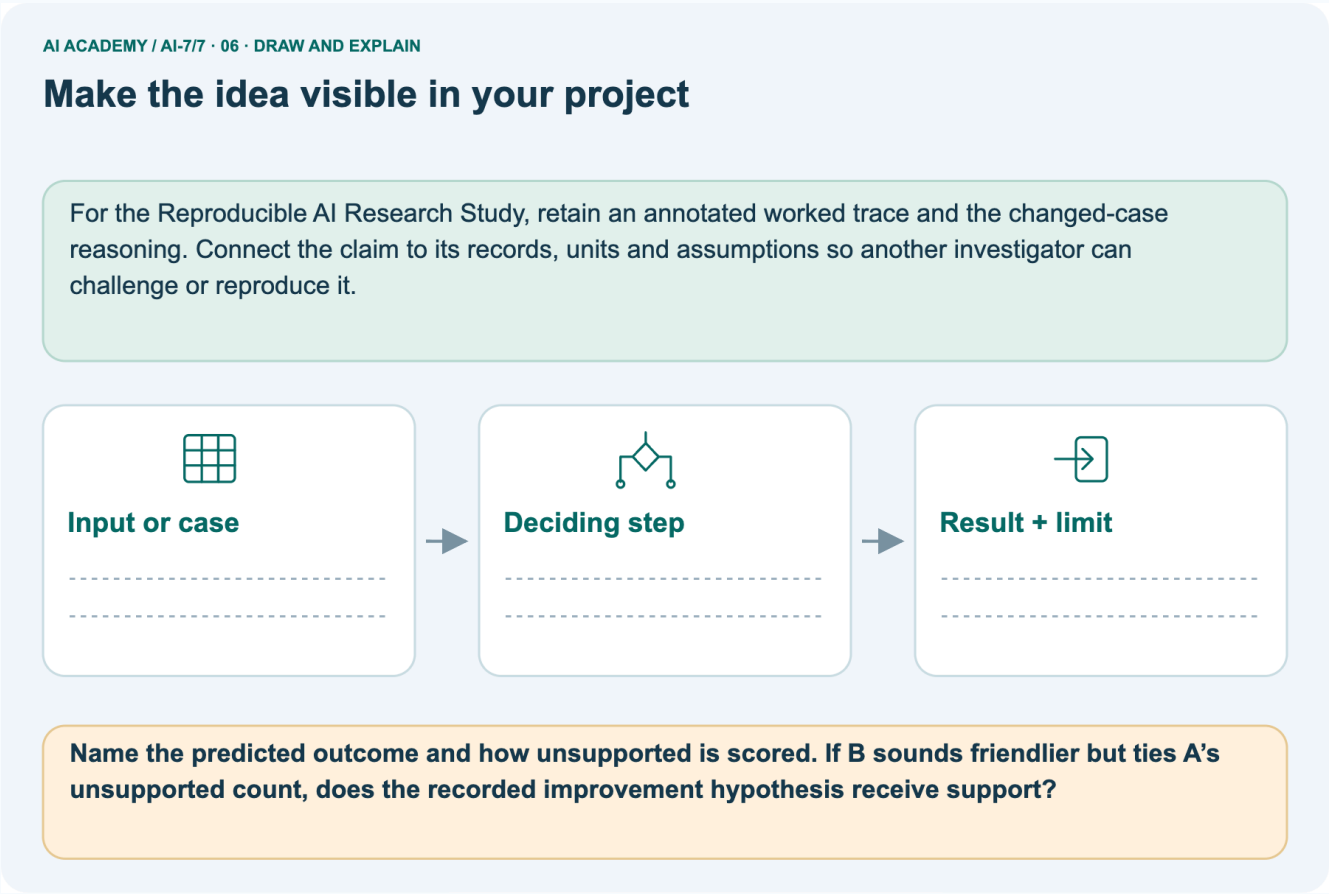
A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Limit: No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Hypotheses and measurable outcomes: the case

02. Follow the deciding steps

03. Hypotheses and measurable outcomes — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.